

Alexander Helbok*, Max Koppelstätter†

January 16, 2025

Abstract

In this experiment we investigate the properties of amplitude modulated (AM), as well as frequency modulated (FM) signals and later couple them to an acousto-optical modulator (AOM) to imprint the message onto a **laser**. We start by investigating the effect of varying the modulation index on the spectrum for AM and FM signals. For AM we notice an increase of sideband width with increasing modulation index, whilst for FM signals we can predict the sideband peak height using Bessel functions. Next we couple an AM signal to the AOM and record the input signal, as well as the laser intensity of the first diffraction order. Using this we create Bode-plots of the gain and phase response of the AOM. The AOM acts like a higher order low pass filter with cutoff frequency in the MHz range. Lastly we drive the AOM with a square pulse and use the time delay and flank broadening of the output signal to estimate the cutoff frequency and phase shift. We again get a cutoff frequency in the MHz regime and a phase shift $\sim \pi$, both of which is in accordance with expectations.

*alexander.helbok@student.uibk.ac.at

†max.koppelstaetter@student.uibk.ac.at

Contents

0	Introduction	1
1	Theory	1
1.1	Modulation	1
1.2	Amplitude modulation	1
1.3	Frequency modulation	2
1.4	Acousto-Optical modulator	3
1.5	Transfer function	3
2	Setup	4
3	Results	4
4	Discussion	9

0 Introduction

Modulation describes different techniques which aim to transfer information onto a carrier signal which is in a different frequency range than the original signal. It is the basis of telecommunication, enabling one to bring radio and TV signal to everybody's home. The idea of all these modulation techniques is to encode the original signal in a carrier signal of a much higher frequency, by varying amplitude, frequency or phase. Due to the type of encoding the information, the three types are named phase, frequency or phase modulation.

In this experiment, the techniques for modulating the frequencies and amplitudes are further investigated. The experiment is mainly split in two parts. In the first part modulated signals are created with a frequency generator. A number of amplitude and phase modulated signals with different modulation indices are recorded. Then, a setup using a HeNe laser is realized, where the beam is transversing an acousto-optical modulator. An amplitude modulated radio-frequency drives the AOM. The first-order diffraction gets recorded. Using this setup, A Bode diagram is measured.

The report starts with a brief theoretical input about modulation techniques and the AOM, then describes the setup. The next section presents the results of the measurements, followed by a discussion.

1 Theory

This section presents the relevant theory for the experiment, covering the used modulation techniques as well as information about the acousto-optical modulator and transfer functions, which are used in the analysis.

1.1 Modulation

Modulation refers to the practice of imprinting information onto a carrier signal. In order to do this a certain property of the carrier is modified according to the message to be transmitted. We choose our carrier signal to be a sinusoidal wave of the form $A \cos(2\pi ft + \phi)$, with amplitude A , frequency f and phase ϕ . These three variables can be varied to modulate the carrier signal and imprint information.

1.2 Amplitude modulation

As the name suggests in amplitude modulation the amplitude of the carrier signal is modulated. The most general form of an amplitude modulated (AM) signal is described by

$$x(t) = A(C + m(t)) \cos(2\pi f_c t), \quad (1)$$

with a constant amplitude scaling A , a constant C chosen to be always larger than the message signal $m(t)$ and the carrier frequency f_c . The modulation index μ of a modulated signal refers

to the ratio between the maximum value the modulated variable takes to the “normal” one. In the case of amplitude modulation the modulation index is

$$\mu = \frac{\max(|m(t)|)}{C}. \quad (2)$$

Using Fourier analysis we can predict how the spectrum of an AM-signal looks like. For sinusoidal message signals with frequency f_m , at most three peaks are expected, located at the carrier frequency f_c and at $f_c \pm f_m$ called the lower and upper sideband (depending on the type of AM modulation some peaks may be suppressed). An AM signal with corresponding spectrum can be seen in Figure 1.

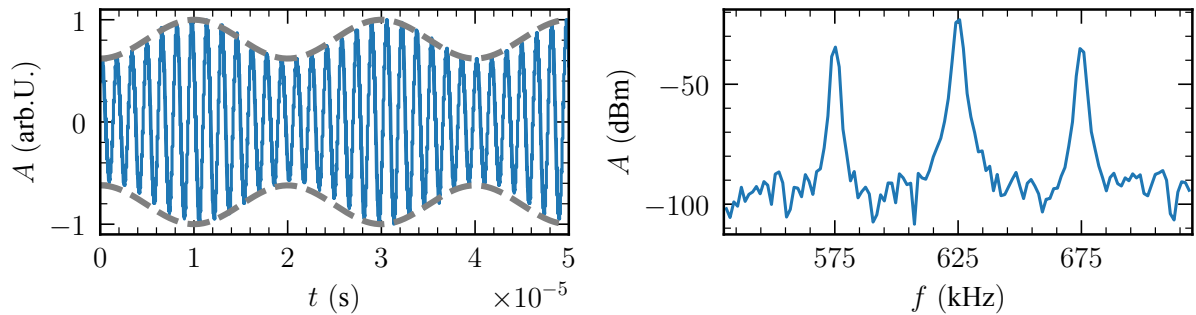


Figure 1: On the left an amplitude modulated signal with a carrier frequency of $f_c = 625$ kHz and modulation frequency of $f_m = 50$ kHz is shown in blue. The enveloping sinusoidal is depicted as dashed gray line. On the right the spectrum of the AM signal is shown. Three peaks are clearly visible located at $f_c, f_c \pm f_m$ corresponding to the carrier frequency and the two sidebands.

1.3 Frequency modulation

Apart from the amplitude we can also modulate the argument of the sine, also known as angle modulation. We will not be modulating the phase so the theoretical description will limit itself to modulation the frequency, the interested reader may refer to [1] for more information on phase modulation.

A frequency modulated (FM) signal can be expressed as

$$x(t) = \cos \left(2\pi f_c t + a \int m(t) dt \right), \quad (3)$$

with carrier frequency f_c and message signal $m(t)$. The modulation index of a FM signal is given by the ratio between the peak frequency deviation to the bandwidth W of the message, that is

$$\mu = \frac{a}{2\pi W} \max(|m(t)|). \quad (4)$$

The spectrum of angle modulated signals is hard to describe on a general basis so we will consider sinusoidal message signals with modulation frequency f_m . Again using Fourier analysis

one can decompose the FM signal into an infinite sum

$$\sum_{n=-\infty}^{\infty} \frac{AJ_n(\mu)}{2} [\delta(f + f_c + nf_m) + \delta(f - f_c - nf_m)]. \quad (5)$$

peaks centered symmetrically around the carrier frequency f_c and spaced by the modulation frequency f_m . The amplitude of the peaks are given by Bessel function $J_n(\mu)$ of order n . Even though the sum is infinite the Bessel functions quickly fall off, with a higher modulation index giving rise to more peaks. A FM signal with modulation index $\mu = 3$ and corresponding spectrum can be seen in Figure 2.

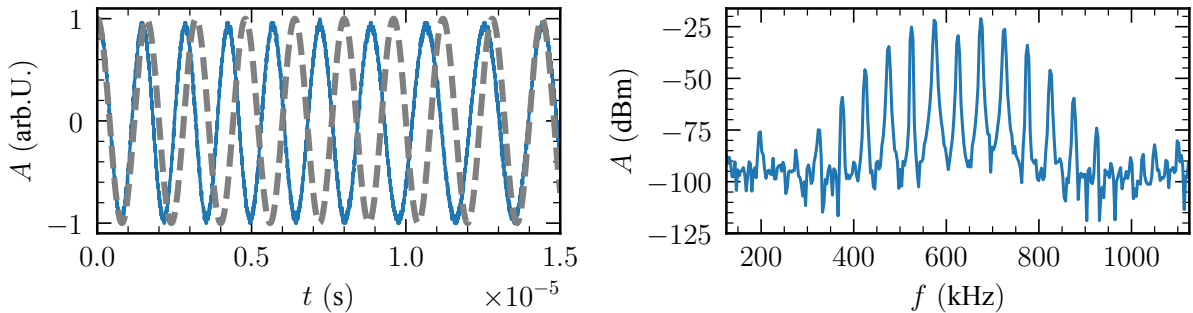


Figure 2: On the left a frequency modulated signal with a carrier frequency of $f_c = 625$ kHz and modulation frequency of $f_m = 100$ kHz is shown in blue. The unmodulated carrier is depicted as dashed gray line. On the right the spectrum of the FM signal is shown with six sidebands clearly visible.

1.4 Acousto-Optical modulator

For the sake of brevity we will keep the explanation of the acousto-optical modulator (AOM) used in this experiment short. An AOM consists of a crystal (TeO_2 in our case) attached to a piezoelectric actuator, which can be driven/ramped using a function generator. Doing this induces local density fluctuations that propagate through the crystal and follow the shape of the driving signal. Since the refractive index of a crystal depends on the density, light shining through an AOM experience time dependent diffraction. In this way we can imprint a message onto a laser.

1.5 Transfer function

We will give a brief introduction to the transfer function since it will be used in the last part of the experiment. The transfer function of a system characterizes how a system reacts to a certain input. It is defined as the ratio between the Laplace transforms of output $y(t)$ and input $x(t)$ signal, as $H(s) = \frac{\mathcal{L}\{y(t)\}}{\mathcal{L}\{x(t)\}}$. For n^{th} -order Butterworth filters the transfer function looks the following way [2]:

$$H(s = jf) = \frac{1}{\sqrt{1 + \epsilon^2 \left(\frac{f}{f_c}\right)^{2n}}}, \quad (6)$$

with the cutoff frequency f_c , the pass band gain ε and the filter order n . f_c represents the frequency at which the filter starts dampening (usually defined as the frequency at which the gain is reduced by 3 dB). ε takes on a value of 1 if the cutoff frequency is defines at the -3 dB point following $f_{-3 \text{ dB}} = f_c \sqrt{2^{1/n} - 1}$. The order of the filter is a measure for the dampening strengths, with higher values of n coming closer to a true rectangular filter.

2 Setup

For the first set of measurements, a frequency generator is used, which creates modulated signals, and is connected to an oscilloscope. The oscilloscope can either record and save the time trace of a signal, but is also able to do a Fast Fourier Transformation. Thus, for some measurements, the FFTs of the traces are recorded. The discrete settings and measurements are presented alongside the results.

For the next measurements, the setup shown in Figure 3 is realized. The signal generator is now again connected to an oscilloscope as well as to a radiofrequency (RF) generator at 80 MHz. Then, the generator is connected to an RF amplifier, which is connected to an acousto-optical modulator (AOM). A HeNe laser beam is redirected by two mirrors, until it enters the AOM. The 0th diffraction order has to be blocked, but the 1st order is recorded by the oscilloscope. Now, a Bode diagram is measured. Here, always input and output signals are recorded. After the Bode diagram, the signal shape is changed to a rectangular signal and traces are saved.

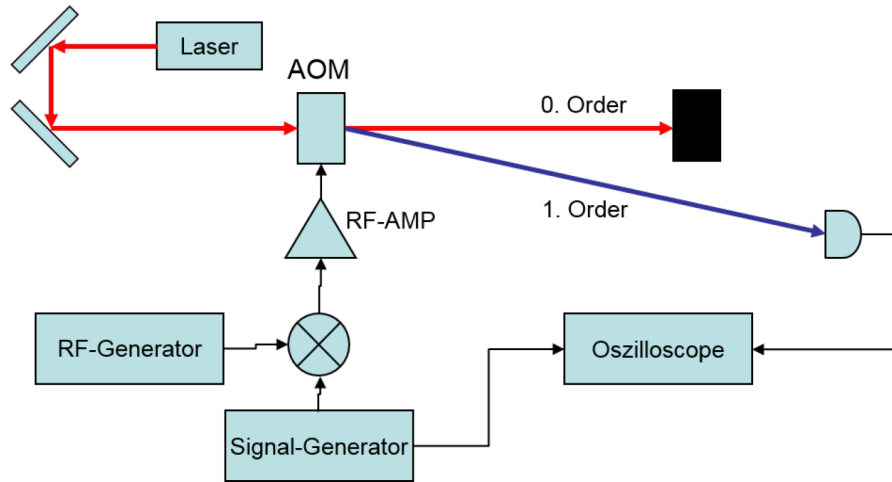


Figure 3: Schematic picture of the setup of the experiment. The AOM is coupled to a modulated signal created by a function generator and amplified by a RF-Amp. Laser from a HeNe Laser is passed through the AOM and intensity of the first diffraction order is measured using a photodiode. Both, input and photodiode signal are recorded on an oscilloscope. Figure taken from [1].

3 Results

We start by studying amplitude modulated (AM) signals with carrier frequency $f_c = 625$ kHz and modulation frequency $f_m = 50$ kHz created by a signal generator. We record time-traces

as well as spectra (using the built-in fast-Fourier transform function of the oscilloscope) for modulation indices of 10 %, 50 % and 100 %. A recorded time-trace can be seen in Figure 1 but our analysis will focus on the spectra of the modulated signals since interesting properties like the bandwidth modulation frequency are encoded in it. In Figure 4 the FFT spectra for the three modulation indices are displayed. A fit consisting of the sum of three independent Lorentz functions was performed and can be seen as solid line.

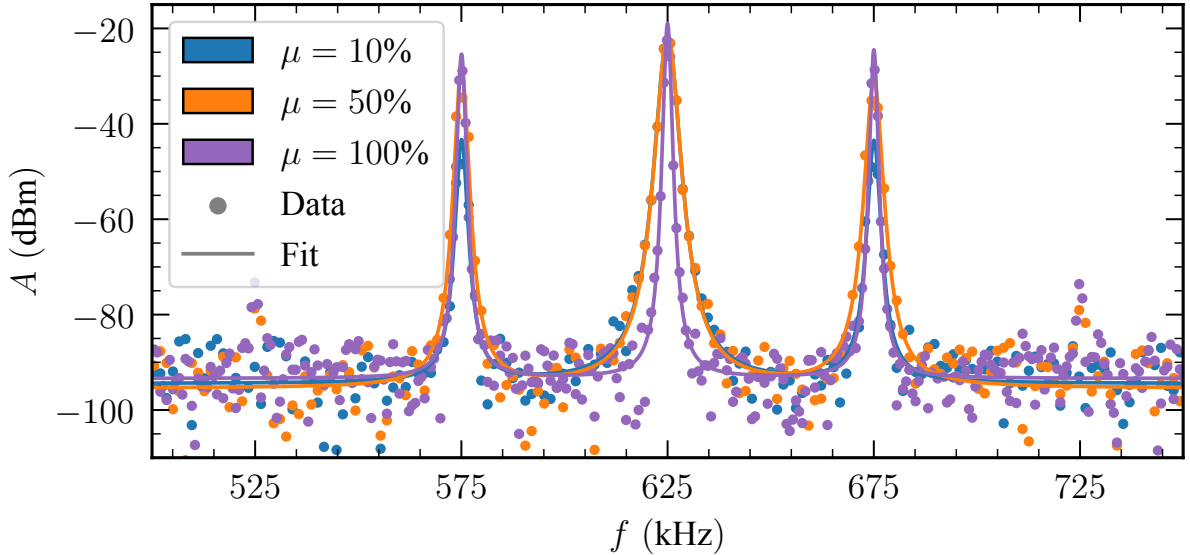


Figure 4

We can clearly identify three distinct peaks for all values of μ ; the central one corresponding to the carrier signal and the outer ones being the lower and upper sideband at $f_c \pm f_m$. From qualitatively looking at the data one can notice two different effects that alter the peak height and width for different modulation indices. Going from 10 % to 20 % leaves the central peak unaltered (as the modulation does not influence the carrier Fourier decomposition). When going to $\mu = 100$ % however the width of the peak is reduced. This can be traced back to the FFT-bandwidth used to create the spectrum, which was changed for $\mu = 100$ %. For the sidebands we first see an increase in peak width as the modulation index is increased, which is then compensated for by reducing the FFT-bandwidth. Furthermore the peak height of the sidebands increases with μ .

We proceed by creating and analyzing FM signals. Again we use a carrier frequency of 625 kHz and adjust the modulation signal to vary the modulation index. An FM time-trace with $f_m = 100$ kHz and $\mu = 3$ with corresponding spectrum is shown in Figure 2. The time-trace is overlaid with the unaltered carrier signal to highlight the frequency deviations induced by the modulation signal. Next we analyze the sideband peak height with respect to modulation index. For this we record 13 FFT spectra and use a combination of peak-detection algorithms and Lorentz fits to automatically extract the peak height of the peaks. This is shown in the top plot of Figure 5. According to Equation (5) the relative peak heights can be related to the Bessel functions argument at the respective modulation index. In the three lower plots of Figure 5 the fitted peak heights with corresponding Bessel function for three different sidebands ($n = 0, 2, 7$)

are exemplary shown.

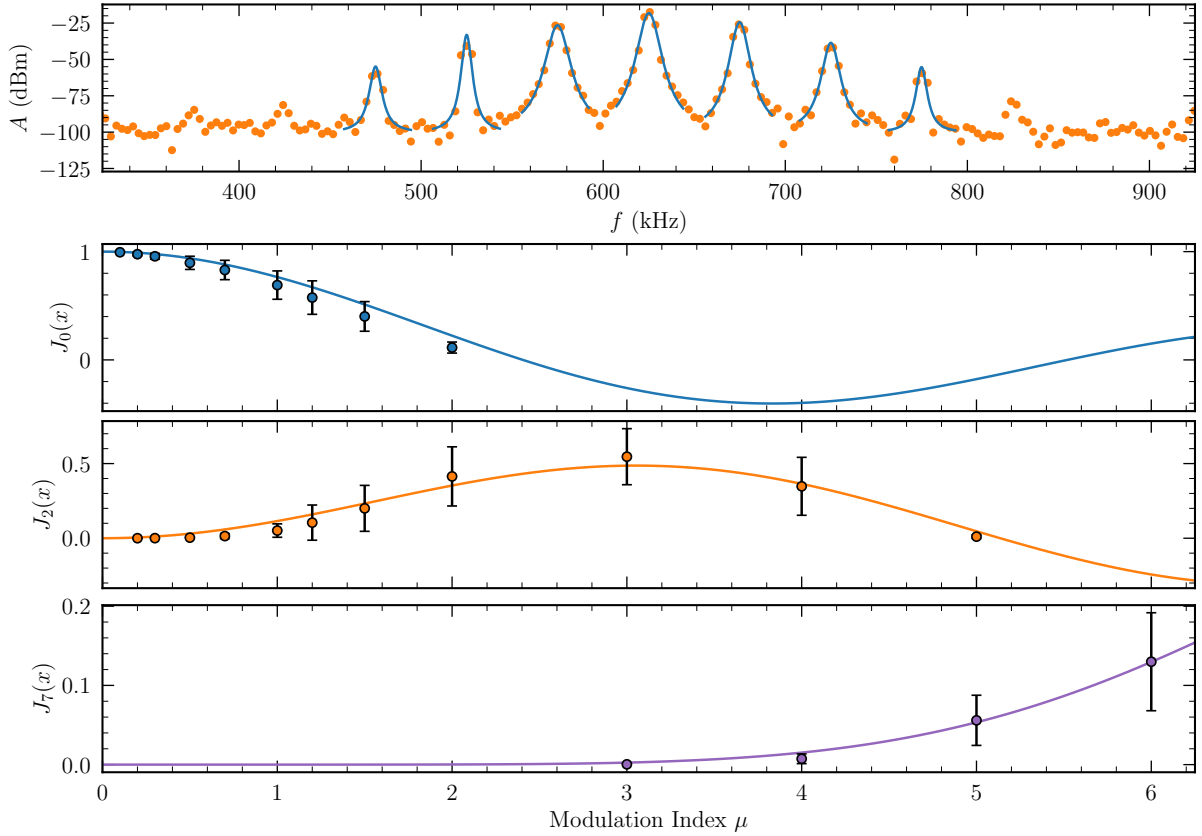


Figure 5: In the top plot Lorentz profiles are fitted to noticeable sidebands in the FM spectrum. By tracking various sidebands across different modulation indices we can plot the relative sideband peak amplitude against the modulation index and compare it to the Bessel function of corresponding order. The datapoints are determined using automatic Lorentz fits with the FWHM of the Lorentz profile chosen as uncertainty. Only parts of the Bessel function for which the amplitude does not get close or below zero can be tracked, as the noise-floor starts to dominate.

The three sidebands depicted were chosen to point out a few things. The central peak (e.g. $n = 0$) can only be traced until $\mu \approx 2$, at which point the relative peak amplitude becomes negative and gets drowned by the noise-floor. For sidebands that remain positive and differ noticeably from zero we can track almost across the whole measurement range, as can be seen for $n = 2$. Higher sidebands (e.g. $n = 7$) only become apparent for high modulation indices. The FWHM of the fitted Lorentz profile was chosen as uncertainty of the datapoints and varies noticeably from datapoint to datapoint. This could be an artifact of poor fits performed by the automatic peak fitting routine and has to be further investigated.

After investigating basic properties of modulated signals we proceed by imprinting the modulated signal onto a laser beam via an acousto-optical modulator. To gain insights into the working characteristics of the AOM a Bode-Diagram is recorded by saving time-traces of the sinusoidal input signal and measuring the laser beam intensity using a photodiode. We start at a modulation frequency of 2 kHz and go up to 10 MHz applying a logarithmic spacing. We extract amplitude and phase for all waveforms by fitting and calculate the signal attenuation

and phase shift between input and output signal. These quantities can be seen in Figure 6 as a function of frequency. The gain profile was fitted to using an n th order bandpass filter function described in Equation (6), whilst the phase response can be described using [3]

$$\phi(f) = \arctan \left[\frac{1}{\alpha} \left(2 \frac{f}{f_c} + \sqrt{4 - \alpha^2} \right) \right] - \arctan \left[\frac{1}{\alpha} \left(2 \frac{f}{f_c} - \sqrt{4 - \alpha^2} \right) \right]. \quad (7)$$

Here the dampening ratio α and cutoff frequency f_c are used for fitting. The last two datapoints were not included in the phase fit since they do not decrease as expected but instead plateau off.

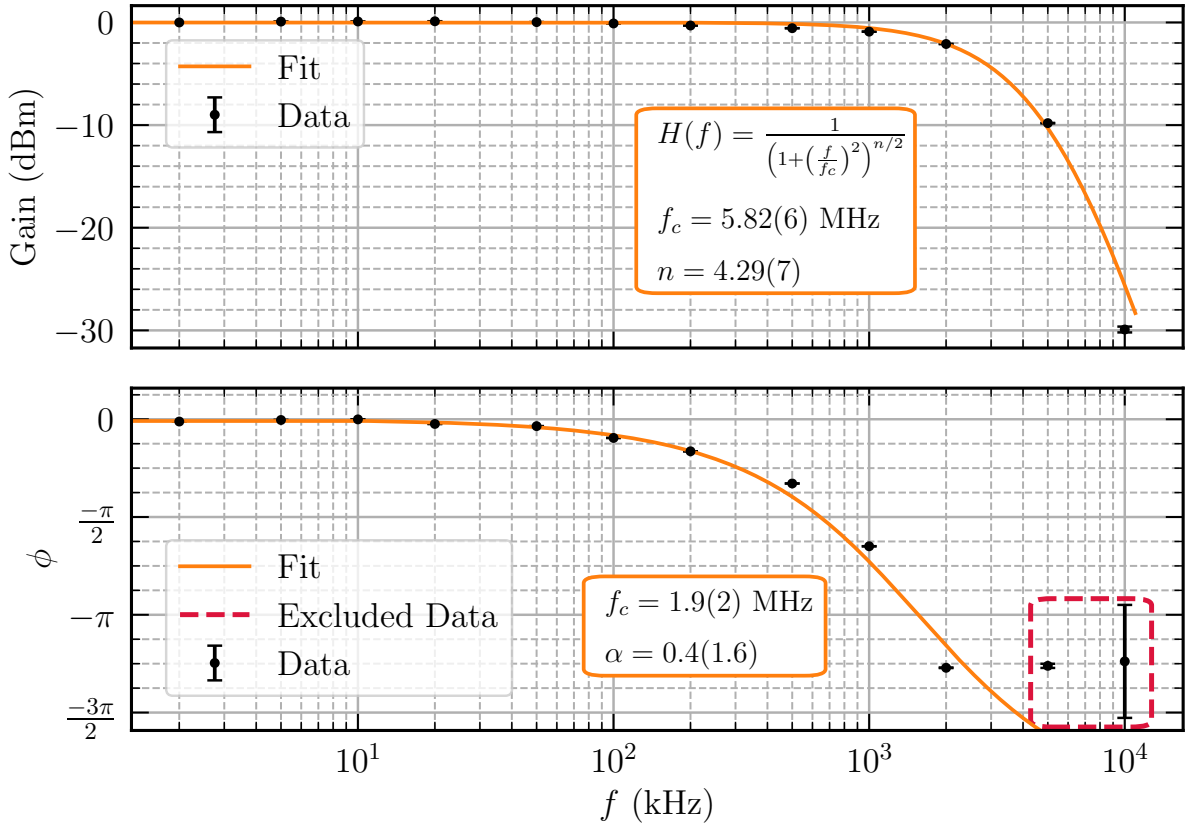


Figure 6: The Gain and phase response of the AOM is plotted against frequency. A n^{th} order bandpass filter is fitted to the gain profile, whilst a function consisting of arctan functions was fitted to the phase shift, both denoted by solid lines. The last two datapoints in the lower plot were excluded from the fit.

From Figure 6 we can notice that the AOM acts like a low pass filter with a cutoff frequency in the range of MHz, e.g. a bandwidth in the range of MHz (the exact value depends how you define the cutoff frequency). Fitting the transfer function to the gain profile leads to a cutoff frequency of $f_c = 5.82(6)$ MHz and a filter order of $n = 4.29(7)$, whilst the phase fit leads to a cutoff frequency of $f_c = 1.9(2)$ MHz. These values are not consistent with one another. The cause for this is unclear but a few things can be pointed out. First, a finer datapoint spacing above the cutoff is desirable for the gain profile as these datapoints determine the two fit parameters (at this point they are somewhat correlated making an exact determination hard). Similarly a finer spacing would give us more insights into the plateau region allowing us to better

understand the behavior by tracking the sinusoidal output.

Multiple factors can limit the bandwidth of the AOM. Firstly, the finite transit time of the acoustic waves traveling through the AOM place an upper bound on the bandwidth. The piezo transducer that induces the density waves could have a cutoff frequency after which it can not keep up with the driving signal. The phase shift of the output signal compared to the input can be understood on similar terms. The finite speed of the acoustic waves lead to a time delay as they need some time to reach and interact with the laser beam. Furthermore the response time of the AOM adds to the phase shift.

Lastly we drive the AOM with a square pulse and record both input and output signal. The recorded data can be seen in Figure 7. To extract the switching time of both signals, as well as the time delay between them we proceeded the following way. We start by calculating the mean and standard deviation in the plateau regions (marked in red in Figure 7). Then we look for the first datapoint that deviates more than 4σ from the mean, since this is highly unlikely ($\approx 0.03\%$) to happen by chance. The uncertainty of these points was estimated using the spacing of the points.

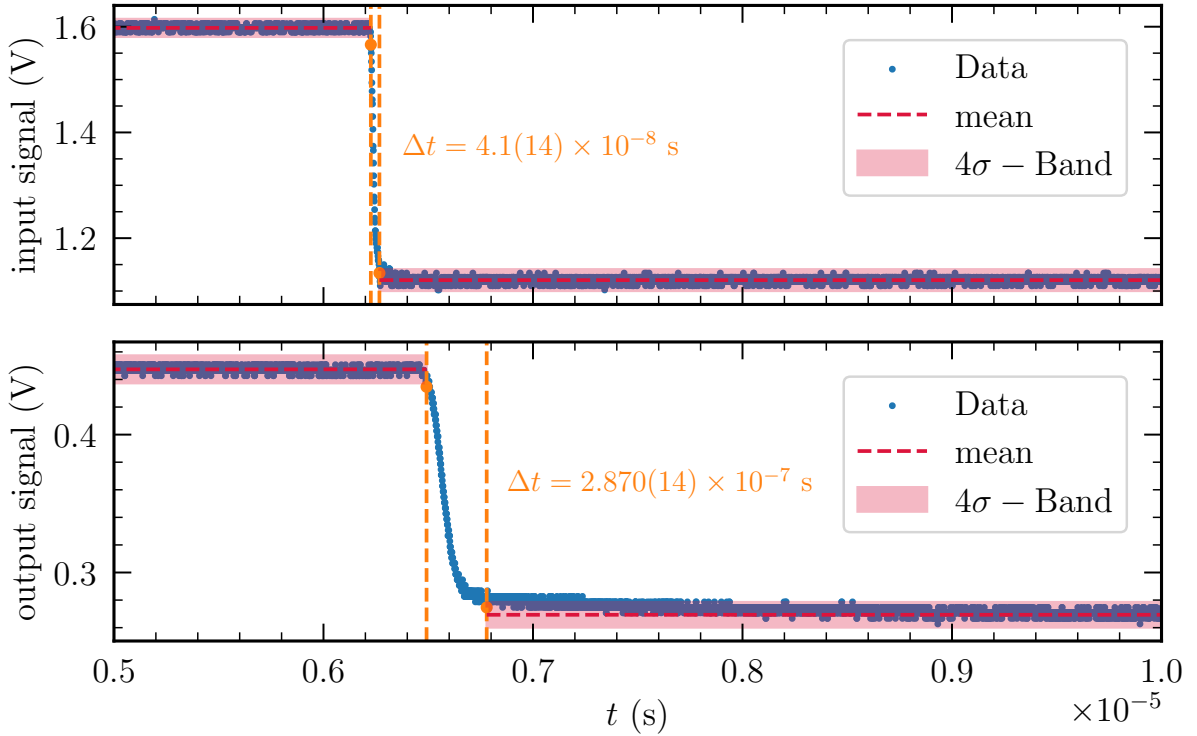


Figure 7: The square pulse driving the AOM can be seen in the top plot, whilst the measured laser intensity after the AOM is plotted in the lower diagram. Using the calculated mean and standard deviation in the plateau regions (red) we can extract the switching time and time delay of the two signals, denoted by dashed orange lines.

Looking at Figure 7 a noticeable difference between input and output signal can be made out. First of all, the switching time of the AOM is slower by an order of magnitude. However the detection of the signal-end datapoint in the output signal is lacking since the constant approximation used to detect the signal is not appropriate; approximation using a linear relation could improve this. We can relate the switching time extracted from the output signal to the

cutoff frequency of the AOM via $f_c = 1/2\Delta t = 1.742(9)$ MHz (we double to get a full switching cycle instead of half). This value lies in the range of the cutoff frequencies estimated in Figure 6 but is likely to be larger, depending on the criteria used to determine the switching time.

Instead what we can determine with good fidelity is the time delay between the two signals using the start of the signal. We get $\delta t = 2.660(14) \times 10^{-7}$ s. This time delay is what causes the phase shift observed in the Bode-plot. We can calculate the phase shift using $\Delta\phi = 2\pi f_c \delta = \pi \delta t / \Delta t = 2.91(3)$, using the cutoff frequency calculated from the switching time. This lies close to a phase shift at the cutoff frequency of π , which is in line with a second order lowpass filter.

4 Discussion

We start by qualitatively investigating amplitude and frequency modulated signals created by a function generator. For the sake of simplicity we use sinusoidal carrier and modulation signals with frequencies in the kHz range. We record frequency spectra for AM signals with varying modulation index and notice a broadening of sidebands with increased modulation index. This is expected as the bandwidth is increased, broadening the sidebands.

The sideband peaks of FM signals with increasing modulation index is described by Bessel functions. To verify this relation we record spectra for modulation index ranging from 0.1 to 6 and using automatic Lorentz fits to extract the peak height and width of all sidebands. We can successfully recreate the trend described by the positive part of the Bessel functions. Negative peak amplitudes can not be measured since they are drowned in the noise floor.

We proceed by creating an AM signal and coupling to an acousto-optical modulator via an RF amplifier and piezo transducer. Laser light from a HeNe-Laser is shone through the AOM to imprinting the modulated signal onto the laser. We measure the input AM signal as well as the laser intensity after passing through the AOM using a photodiode to capture the gain and phase characteristics of the AOM in a Bode-plot. We fit an n^{th} order Butterworth filter transfer function to the response plots and extract the cutoff frequency of the lowpass filter. Fitting to gain yields $f_c = 5.82(6)$ MHz, with the phase shift leading to $f_c = 1.9(2)$ MHz. These values are not consistent with each other. A finer frequency spacing has to be chosen close to and above the cutoff frequency, since these datapoints are important for fitting.

Lastly we drive the AOM with a square pulse and measure the response of the system by recording input and output signal on the oscilloscope. We notice an increase in switching time (e.g. a stretching of the falling flank of the square wave) by an order of magnitude, as well as a time delay between the two signals. These two quantities let us estimate the cutoff frequency, as well as the phase shift at f_c , leading to $f_c = 1.742(9)$ MHz and $\Delta\phi = 2.91(3)$. The phase shift matches expectations for a second order lowpass filter, with the f_c lying in the range of the values extracted from the Bode-plot. However more precise measurements are needed to get a better and more importantly consistent estimate for the cutoff frequency of the AOM.

References

- [1] *Exercise FP2-A-04: Modulation*. Universität Innsbruck, Innsbruck, AT: Institut für Experimentalphysik.
- [2] *Butterworth Filter Design*. Online; Accessed on the 13. 01. 2025. URL: https://www.electronics-tutorials.ws/filter/filter_8.html.
- [3] H. Zumbahlen. *Phase Relations in Active Filters*. Online; Accessed on the 13. 01. 2025. URL: <https://www.analog.com/en/resources/analog-dialogue/articles/phase-relations-in-active-filters.html>.

Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



.....
Alexander Helbok

.....
January 16, 2025

.....
Date



.....
Max Koppelstätter

.....
January 16, 2025

.....
Date